

injury, and only 2 teams had 2 injured players each. Rain was a factor in only one match. Of the injuries that occurred during matches, 5 happened in the first half and 12 in the second half. The right knee was affected in 14 players. In 12 cases (54%), the injured knee was on the player's dominant side. The average height was 1.78 meters. The positions of the injured players were: 9 forwards, 7 midfielders, and 6 defenders. All players wore boots with a lateral stud configuration, and 3 different brands were identified. Based on video analysis, 31% of injuries were caused by a direct mechanism, 23% by indirect contact, and 46% occurred without contact. Conclusion: This homogeneous cohort of professional league soccer players underscores a concerning increase in the incidence of ACL injuries over a short period. Our study identified several factors contributing to these ACL ruptures, with no single determinant emerging as the primary cause. The significant rise in injury rates highlights the severity of the problem and emphasizes the urgent need for intervention. We propose that the main contributing factors include the high frequency of matches played at excessive intensity, reduced training time, limited focus on injury prevention, and inadequate rest periods. Addressing these factors is crucial for mitigating the risk and severity of ACL injuries.

Category: Knee - ACL

No Detectable Changes in Dynamic Knee Kinematics During Downhill Running 10+ Years After Anatomic ACL Reconstruction

Abstract ID# 24939

All Authors:

Anja M Wackerle MD GERMANY
Tom Gale MS UNITED STATES
Koji Nukuto MD, PhD UNITED STATES
Stephen J. Rabuck MD UNITED STATES
William Anderst PhD UNITED STATES
James J. Irrgang PT, PhD, FAPTA UNITED STATES
Volker Musahl MD, Prof. UNITED STATES

Disclosure information for authors is available at <https://isakos.com/2025/Papers>.

Summary:

No detectable change in dynamic knee kinematics was found between short-term and long-term follow-up using biplane radiography during downhill running 10+ years after anatomic anterior cruciate ligament reconstruction and no difference between limbs could be detected at 10+ years postoperative.

Data:

Introduction: Substantial evidence from biomechanical studies shows that anterior cruciate ligament (ACL) reconstruction (ACLR) improves anterior-posterior and rotatory stability. However, previous in-vivo studies have reported that ACLR fails to fully restore physiological kinematics within 24 months, leaving unclear how knee kinematics are affected in the long-term. This study aims to investigate knee kinematics after anatomic ACLR with quadriceps autograft, utilizing dynamic biplane radiography at a minimum of 10-year follow-up. It was hypothesized that knee kinematics remain unchanged between short- and long-term follow-up after ACLR and that the surgical knee 10+ years after ACLR has similar kinematics compared to the contralateral healthy knee. Methods: This study enrolled patients from a previous randomized controlled trial in which patients underwent primary anatomic single- or double-bundle ACLR with a quadriceps tendon (QT) autograft. In vivo knee kinematics were previously measured at 6 and 24 months postoperatively. Exclusion criteria for the current study included ACL revision surgery or knee arthroplasty. At a minimum of 10-years follow-up, synchronized biplane radiographs of both knees were obtained in the same laboratory. Participants performed downhill running at 3 m/s. A dual belt instrumented treadmill recorded ground reaction forces at 900 Hz with a threshold of 50N to identify foot strike and toe off. Volumetric and surface models of the femur and tibia were obtained previously by segmentation of CT scans (Mimics, Materialize, Inc.). A validated model-based tracking system was used for three-dimensional (3D) positioning and orientation of the tibia and femur, allowing the calculation of 3D motions of the tibia and femur. Analyzed kinematic parameters were tibiofemoral flexion, internal rotation, abduction, and anterior translation, analyzed at 0%, 20%, and 40% of the stance phase. Linear mixed effects models were created to compare joint kinematics at 6 months, 24 months, and 10+ years and between limbs within each individual,

with significance set at $p < 0.05$. Clinical outcome parameters collected included the International Knee Documentation Committee Subjective Knee Form (IKDC SKF). Results: Seven subjects of the initial randomized controlled trial have been tested (mean age at surgery 27.4 ± 10.5 years; 1 (14%) female; 1 (14%) single-bundle ACLR) at a follow-up of 12.5 ± 0.6 years. None of the assessed kinematic parameters within subject showed significant differences among the three time points in the surgical knee. No significant differences were identified between surgical and contralateral limb kinematics at 10+ years after surgery. Mean IKDC SKF at the latest follow-up was 85.7 ± 14.7 . Conclusion: There was no detectable change in knee kinematics after anatomic ACLR from short-term to long-term follow-up in this cohort of seven patients. Additionally, there were no differences in knee kinematics between the surgical and contralateral knee at long-term follow up. However, the small sample size is not sufficient to statistically conclude that knee kinematics are unchanged over time and results are limited to subjects without ACLR revision. Enlarging the sample size bears the potential to further assess whether anatomic ACLR effectively restores and maintains joint kinematics similar to the native contralateral limb over the long-term.

Category: Knee - ACL

Adductor Canal Block in Anterior Cruciate Ligament Reconstruction Surgery: A Prospective, Randomized Controlled Trial

Abstract ID# 25707

All Authors:

Connor Delman MD UNITED STATES
Shannon Tse BMBS UNITED STATES
Augustine Saiz MD UNITED STATES
Trevor Shelton MD, MS UNITED STATES
Matthew Pena MD UNITED STATES
Cassandra Lee MD UNITED STATES

Disclosure information for authors is available at <https://isakos.com/2025/Papers>.

Summary:

Preoperative adductor canal block improves immediate postoperative pain in ACL reconstruction patients but offers no long-term benefit for pain control or opioid consumption beyond 24 hours.

Data:

Introduction: There are no consensus guidelines for perioperative pain management in anterior cruciate ligament reconstruction (ACLR) surgery. The adductor canal block (ACB) is a motor-sparing alternative to femoral nerve block, with studies showing no difference in postoperative opioid consumption or pain. Few randomized studies have assessed its efficacy beyond the immediate postoperative period. The aim of this study was to compare the efficacy of a single preoperative ACB versus no block on postoperative pain control following ACLR. We hypothesized that ACB would reduce pain and opioid use without affecting functional outcomes. Methods: This was a prospective, randomized controlled trial of all adult patients undergoing primary ACLR at a single academic institution from 2018-2024. Patients were randomized to receive either ACB or no block (NB) preoperatively. All patients received a standardized preoperative regimen (400mg celecoxib, 300mg gabapentin, and 1000mg acetaminophen). The surgical team was blinded to the treatment intervention. Patients given hydromorphone were excluded from the study. All patients completed the Brief Resilience Scale (BRS) the day of surgery and scores were stratified into low, normal, or high resilience groups. Intraoperative fentanyl morphine equivalent doses (MED), postoperative visual analogue scale (VAS) pain level and MEDs were assessed in the post-anesthesia care unit (PACU) after 15 minutes and daily through 10 days postoperatively. Knee Injury and Osteoarthritis Outcomes Scores (KOOS) were completed at 6 and 12 weeks postoperatively. Results: 50 patients were analyzed (ACB, n=25; NB, n=25), with no differences in demographics or surgical characteristics. The cohort consisted of a young patient population (ACB mean age = 24.6, NB mean age = 23.6, $p=0.39$). The ACB group had a lower mean BRS (3.70 vs 4.11, $p=0.02$), though both groups remained within the "normal resilience" range. Intraoperative fentanyl MEDs were similar (2.48 vs 2.45, $p=0.91$). PACU VAS scores were significantly lower in the ACB group at 15 minutes postoperatively (3.40 vs 5.62, $p<0.01$), but no difference was noted from the time of discharge onwards (3.64 vs 4.24, $p=0.24$).